

SAMUEL MALEC

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Brno, Czech Republic [◇ Open to Remote and Relocation](#)

EDUCATION

M.Sc. in Theoretical Computer Science	2025 – Present
Faculty of Informatics, Masaryk University, Brno	
B.Sc. in Programming and Development	2022 – 2025
Faculty of Informatics, Masaryk University, Brno	

TECHNICAL SKILLS

Compiler	LLVM, MLIR, GraalVM Native Image, SSA, Sea-of-Nodes, QuickJS
Languages	C++, C, Rust, Python, Java, JavaScript
Tools	Git, CMake, Conan

EXPERIENCE

Masaryk University (Oracle Labs collaboration)	2024 – Present
<i>Junior Researcher</i>	<i>Brno</i>

- Designed and implemented a general-purpose abstract interpretation framework for GraalVM Native Image, executing static analyses over Graal's Sea-of-Nodes IR.
- Investigated context-sensitive interprocedural analysis for ahead-of-time compilation, evaluating precision/performance trade-offs on large-scale applications.
- Developing a compilation pipeline targeting Cthulhu, an SSA-based IR for static analysis, including a JavaScript frontend and QuickJS bytecode backend.

SANEZOO	2024 – 2025
<i>C++ Software Developer</i>	<i>Brno</i>

- Developed C++ industrial communication microservices integrating OPC-UA, Modbus, and Step7.
- Optimized real-time camera acquisition using Allied Vision's Vimba SDK.
- Refactored automated testing infrastructure for distributed microservices.

Masaryk University	2023 – Present
<i>Seminar Tutor</i>	<i>Brno</i>

- Teach introductory Python and C programming and mentor first-year students.
- Prepare teaching materials and programming assignments, mentor, public speaking.

OPEN SOURCE PROJECTS

compiler-dungeon — Compiler pipeline for a custom language
<i>C++, SSA, LLVM, MLIR</i> github.com/samuel-malec/compiler-dungeon

- Developing a statically typed programming language with generics, typeclasses, first-class functions and algebraic data types; implemented recursive-descent parsing, bidirectional type inference, semantic analysis and SSA-based intermediate representations.
- Building lowering pipelines toward MLIR/LLVM and experimenting with optimization and static-analysis passes.

TRAINING/LICENCES

ACM Europe Summer School on MLIR — Selected Participant, 2026